

4. Integration

What students need to learn:

Integration of hyperbolic functions and expressions involving them.

Integration of inverse trigonometric and hyperbolic functions.

For example, $\int \operatorname{arsinh} x \, dx$, $\int \arctan x \, dx$.

Integration using hyperbolic and trigonometric substitutions.

To include the integrals of $1/(a^2 + x^2)$, $1/\sqrt{(a^2 - x^2)}$, $1/\sqrt{(a^2 + x^2)}$, $1/\sqrt{(x^2 - a^2)}$.

Use of substitution for integrals involving quadratic surds.

In more complicated cases, substitutions will be given.

The derivation and use of simple reduction formulae.

Students should be able to derive formulae such as

$$nI_n = (n - 1)I_{n-2}, \quad n \geq 2,$$

$$\text{for } I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx,$$

$$I_{n+2} = \frac{2 \sin(n + 1)x}{n + 1} + I_n$$

$$\text{for } I_n = \int \frac{\sin nx}{\sin x} \, dx, \quad n > 0.$$

The calculation of arc length and the area of a surface of revolution.

The equation of the curve may be given in cartesian or parametric form. Equations in polar form will not be set.
